

Fuzzy Boundary (An Embroidery Installation)

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ABSTRACT

This installation provides a hands-on attendee experience with procedural design workflows and programmable machine embroidery that connect various domains including biomimicry, visualization, and product design. Situated within contemporary studio craft practices of digitizing the handcrafts, this work recognizes time-based computation as machine agent for bio-logical pattern generation, and programmable machine embroidery as the realization of structural biomimicry. Critical making is a form of research and practice among contemporary craft artists, and such studio activity is a design methodology employing various material and technical explorations that are by nature complex interdisciplinary relations. Fuzzy Boundary (An Embroidery Installation) represents redefining educational relationships among 21st century industrial design practice, and visualization.

CCS CONCEPTS

• Human-centered computing → Visualization design and evaluation methods; Interaction design; • Applied computing → Fine arts; Interactive learning environments.

KEYWORDS

critical making, craft, biomimicry, embroidery, interaction design

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1 INTRODUCTION

Fuzzy Boundary focuses on understanding creativity and computation as an expression of the human-machine relationship. This work addresses the generative design algorithm as a form of materiality by assigning attributes of time, actions, and variables to a tangible outcome of textures, and shapes. This project utilizes machine embroidery as the mechanism for prototyping various biomimicry processes and structures. The two-dimensional quality of the embroidery as an abstract graphic or pictograph, is often a cross-section representation of three-dimensional computational reactions. The embroidery output is a data materialization prototype, and a type of graphical mapping system that embodies the

syntax and nature of computational growth systems.[Starrett et al. 2018] Other computational craft projects that model educational opportunities illustrate the advantages of teaching code through a craft tradition.[Smith 2017][Wolz et al. 2019][Lee and Llach 2020] These research models position coding pedagogy as a means of translating handwork through visual computing. The proposed project works with the same objective as Lee and Llach, a design methodology and technology system that produces tangible embroidery paths and machine code.[Lee and Llach 2020] Though unlike the computer feedback loop introduced by Lee and Llach, Fuzzy Boundary models a workflow for making embellished textiles and products.



Figure 1: An example of an embroidered grid pattern and vector field.

2 THE BIO-MIMICRY DESIGN SPACE

The tools used for this generative design project are SideFX Houdini and pyembroidery (<https://github.com/EmbroiderPy/pyembroidery>) a Python library for the reading and writing of embroidery files. Attendees will be able to interact with generative scripts within Houdini that visualize fundamental geometry, fractal systems, and eight structural biological elements common among animal taxonomy. According to Naleway et. al, fibrous, helical, gradient, layered,

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tubular, cellular, suture, and overlapping are attributes identified as biological structures for explaining complex biological materials.[Naleway et al. 2015] Fuzzy Boundary takes inspiration from these structural attributes and synthesizes them into variable and interactive tools developed within the Houdini environment taking advantage of growth algorithms based on time. L-systems, VDBs, particle systems and vector fields are employed in the development of this set of bioinspired design tools. From the translation of design to machine code, pyembroidery compiles the machine code that drives the programmable embroidery machine. A Python node within Houdini calls the pyembroidery library that reads the ordered points based on connectivity, and drives the positioning of the embroidery needle. The pyembroidery library offers the conference attendee a unique application of computer aided embellishment with machine level controls.

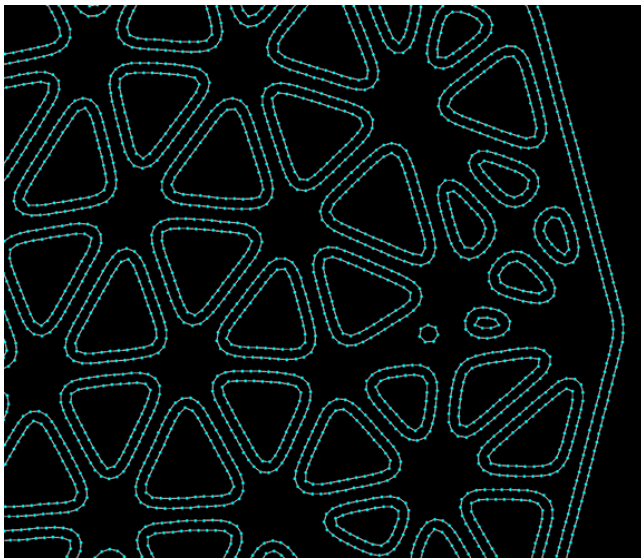


Figure 2: Cross section of cell automata illustrating structured sets of points. These points drive the positioning of the embroidery needle.

2.1 The SIGGRAPH attendee experience

The Labs installation Fuzzy Boundary will give the SIGGRAPH conference attendee an opportunity to manipulate and configure node-based design work-flows within Houdini for the purpose of programming an embroidery machine and fabricating their design. The attendee will take away a 4"x4" embroidered swatch. Attendees do not need prior knowledge of Houdini, nor machine embroidery. The node-based workflows will accommodate beginners as well as the experienced Houdini operator

3 CITATIONS AND BIBLIOGRAPHIES

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